

For example, m_5 and m_7 lie in two adjacent squares. Variable y is primed in m_5 and unprimed in m_7 , while the other two variables are the same in both squares. From the postulates of Boolean algebra, it follows that the sum of two minterms in adjacent squares can be simplified to a single AND term consisting of only two literals.

$$\begin{aligned} m_5 + m_7 &= xy'z + xyz = z \cdot (xy') + z \cdot (xy) \quad (\text{Postulate 3(b)}) \\ &= (zx) \cdot y' + (zx) \cdot y \quad (\text{Theorem 4(b)}) \\ &= (zx) \cdot (y' + y) \quad (\text{Postulate 4(a)}) \\ &= (zx) \cdot (y + y') \quad (\text{Postulate 3(a)}) = (zx) \cdot 1 \quad (\text{Postulate 5(a)}) \\ &= zx \quad (\text{Postulate 2(b)}) = xz \quad (\text{Postulate 3(b)}) \end{aligned}$$

∴ Any two minterms in adjacent squares that are ORed together will cause a removal of the different variable.

Example 3-1: Simplify the Boolean function:

$$F = x'yz + xy'z + xy'z' + xyz$$

	yz			
	00	01	11	10
x			1	1
0				
1	1	1		

Figure 3-4

- First, a 1 is marked in each square as needed to represent the function in Fig 3-4. This can be accomplished in two ways: either by converting each minterm to a binary number and then marking a 1 in the corresponding square, or by obtaining the coincidence of the variables in each term.
- For example, the term $x'yz$ has the corresponding binary number 011 and represents minterm m_3 in square 011.
- The second way to recognize the square is by the coincidence of the variables x', y and z , which is found in the map by observing that x' belongs to the four squares in the first row, y belongs to the four squares in the two right columns, and z belongs to the